

## Relatório iptables

### Projeto 2 - Segurança e Confiabilidade 2025/2026- Grupo 02

#### 1. Política por omissão

Foi aplicada a política por omissão DROP nas chains INPUT, OUTPUT e FORWARD.

A opção DROP segue o princípio de default deny: tudo é bloqueado por omissão e só é permitido o tráfego explicitamente autorizado. Esta foi a escolha mais segura para reduzir a superfície de ataque da máquina.

Foram mantidas duas exceções obrigatórias:

- tráfego de loopback (lo);
- tráfego de ligações já estabelecidas ou relacionadas (ESTABLISHED,RELATED).

#### 2. Regras iptables

Foram usadas as seguintes redes e máquinas:

- SperServ: 10.0.0.2
- SperAdm: 10.0.0.3
- CondAdm: 10.0.3.2
- CondServ: 10.0.3.3
- Broker: 10.0.2.4
- sub-rede dos clientes: 10.0.1.0/24
- sub-rede local de SperServ: 10.0.0.0/24

As regras definidas foram:

- permitir ping a SperServ apenas a partir de SperAdm, CondAdm e da sub-rede 10.0.1.0/24;
- permitir SpertaClient apenas a partir da sub-rede 10.0.1.0/24;
- permitir ssh a SperServ apenas a partir de Broker e da sub-rede local 10.0.0.0/24;
- permitir ping de saída apenas para 10.0.0.0/24 e Broker, com limite de 4 pings por segundo;
- permitir ssh de saída apenas para CondServ.

Comandos iptables utilizados para implementar a política:

```
sudo iptables -P INPUT DROP
sudo iptables -P OUTPUT DROP
sudo iptables -P FORWARD DROP

sudo iptables -A INPUT -i lo -j ACCEPT
sudo iptables -A OUTPUT -o lo -j ACCEPT
sudo iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
sudo iptables -A OUTPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

sudo iptables -A INPUT -p icmp --icmp-type echo-request -s 10.0.0.3 -j ACCEPT
sudo iptables -A INPUT -p icmp --icmp-type echo-request -s 10.0.3.2 -j ACCEPT
sudo iptables -A INPUT -p icmp --icmp-type echo-request -s 10.0.1.0/24 -j ACCEPT

sudo iptables -A INPUT -p tcp --dport 22 -s 10.0.2.4 -j ACCEPT
sudo iptables -A INPUT -p tcp --dport 22 -s 10.0.0.0/24 -j ACCEPT

sudo iptables -A INPUT -p tcp --dport 22345 -s 10.0.1.0/24 -j ACCEPT

sudo iptables -A OUTPUT -p icmp --icmp-type echo-request -d 10.0.0.0/24 -m limit --limit 4/second --limit-burst 4 -j ACCEPT
sudo iptables -A OUTPUT -p icmp --icmp-type echo-request -d 10.0.2.4 -m limit --limit 4/second --limit-burst 4 -j ACCEPT

sudo iptables -A OUTPUT -p tcp --dport 22 -d 10.0.3.3 -j ACCEPT
```

### 3. Método de teste

Os testes foram executados nas webshells do Mininet-GUI. Os comandos usados foram:

- ping -c 4 <IP>
- java -jar dist/SpertaClient.jar 10.0.0.2:22345 ...
- ssh 10.0.3.3
- sudo iptables -L -v -n

O critério de validação foi simples: os casos autorizados devem funcionar, os casos não autorizados devem falhar e os contadores de iptables devem refletir o tráfego permitido.

### 4. Resultados dos testes

| Teste | Origem   | Comando                      | Resultado esperado | Resultado obtido |
|-------|----------|------------------------------|--------------------|------------------|
| T1    | SperCli1 | ping 10.0.0.2                | Sucesso            | Sucesso          |
| T2    | SperAdm  | ping 10.0.0.2                | Sucesso            | Sucesso          |
| T3    | CondAdm  | ping 10.0.0.2                | Sucesso            | Sucesso          |
| T4    | Outsider | ping 10.0.0.2                | Falha              | Falha            |
| T5    | Broker   | ping 10.0.0.2                | Falha              | Falha            |
| T6    | SperCli1 | SpertaClient -> SpertaServer | Sucesso            | Sucesso          |
| T7    | Outsider | SpertaClient -> SpertaServer | Falha              | Falha            |
| T8    | Broker   | ssh 10.0.0.2                 | Sucesso            | Sucesso          |
| T9    | SperServ | ssh 10.0.3.3                 | Sucesso            | Sucesso          |
| T10   | Outsider | ssh 10.0.0.2                 | Falha              | Falha            |
| T11   | SperServ | ping para sub-rede local     | Sucesso            | Sucesso          |
| T12   | SperServ | ping 10.0.2.4                | Sucesso            | Sucesso          |
| T13   | SperServ | ping <Outsider>              | Falha              | Falha            |
| T14   | SperServ | ssh 10.0.3.3                 | Sucesso            | Sucesso          |
| T15   | SperServ | ssh <SperCli1>               | Falha              | Falha            |

### 5. Capturas de ecrã

T1

```
root@SperCli1:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=63 time=5.90 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=63 time=0.516 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=63 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=63 time=0.038 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3034ms
rtt min/avg/max/mdev = 0.038/1.623/5.900/2.476 ms
```

T3

```
root@CondAdm:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=63 time=1.50 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=63 time=0.278 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=63 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=63 time=0.038 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.038/0.471/1.499/0.600 ms
```

T5

```
root@Broker:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3061ms
```

T7

```
drigo@frutasstore-password src/sperta/certs/client.keystore client-password rod
Erro de comunicacao com o servidor: Connection timed out
```

T2

```
root@SperAdm:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.979 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.124 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.030 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.029 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3039ms
rtt min/avg/max/mdev = 0.029/0.290/0.979/0.399 ms
```

T4

```
root@Outsider:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3066ms
```

T6

```
drigo@frutasstore-password src/sperta/certs/client.keystore client-password rod
OK-ATTEST
OK-USER

Comandos disponiveis:
CREATE <hm> # Criar casa <hm>
ADD <user> <hm> <ss> # Adicionar <user> a casa <hm>, seq <ss>
RD <hm> <ss> # Registrar dispositivo na casa <hm>, seq <ss>
EC <hm> <ss> <int> # Enviar valor <int> ao dispositivo <ss> da casa <hm>
RT <hm> # Receber estados de todos os dispositivos da casa <hm>
RH <hm> <ss> # Receber historico (csv) do dispositivo <ss> da casa <hm>
```

T8

```
root@mininet-yw:~# ssh -X root@10.0.0.2
root@10.0.0.2:~# password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

192 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Fri May 15 03:08:01 2026 from 10.0.0.2
```

T9

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.3.3
root@10.0.3.3:~# password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

192 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Fri May 15 02:38:58 2026 from 10.0.0.2
```

T10

```
root@outsider:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.0.2
ssh: connect to host 10.0.0.2 port 22: Connection timed out
```

T11

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.621 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=1.79 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.530 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.038 ms

--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3055ms
rtt min/avg/max/mdev = 0.038/0.745/1.792/0.643 ms
```

T12

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=63 time=1.69 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=63 time=0.455 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=63 time=0.105 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=63 time=0.039 ms

--- 10.0.2.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3019ms
rtt min/avg/max/mdev = 0.039/0.572/1.692/0.665 ms
```

T13

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.10.0.2
PING 10.10.0.2 (10.10.0.2) 56(84) bytes of data:
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted

--- 10.10.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3079ms
```

T14

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.3.3
root@10.0.3.3:~# password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

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Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

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To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Fri May 15 02:52:01 2026 from 10.0.0.2
```

T15

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.1.3
ssh: connect to host 10.0.1.3 port 22: Connection timed out
```

Resultado de sudo iptables -L -v -n para mostrar os contadores de pacotes:

```
-n @SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ sudo iptables -L -v
Chain INPUT (policy DROP 42 packets, 2616 bytes)
pkts bytes target prot opt in out source destination
0 0 DROP icmp -- * * 10.0.2.0/24 0.0.0.0/0 icmptype 8
0 0 DROP icmp -- * * 10.0.2.4 0.0.0.0/0 icmptype 8
0 0 DROP icmp -- * * 10.0.2.0/24 0.0.0.0/0 icmptype 8
0 0 DROP icmp -- * * 10.0.2.4 0.0.0.0/0 icmptype 8
46 8392 ACCEPT all -- lo * 0.0.0.0/0 0.0.0.0/0
445 54276 ACCEPT all -- * * 0.0.0.0/0 0.0.0.0/0 state RELATED,ESTABLISHED
0 0 ACCEPT icmp -- * * 10.0.0.3 0.0.0.0/0 icmptype 8
0 0 ACCEPT icmp -- * * 10.0.3.2 0.0.0.0/0 icmptype 8
0 0 ACCEPT icmp -- * * 10.0.1.0/24 0.0.0.0/0 icmptype 8
8 480 ACCEPT tcp -- * * 10.0.2.4 0.0.0.0/0 tcp dpt:22
0 0 ACCEPT tcp -- * * 10.0.0.0/24 0.0.0.0/0 tcp dpt:22
5 364 ACCEPT tcp -- * * 10.0.1.0/24 0.0.0.0/0 tcp dpt:22345

Chain FORWARD (policy DROP 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination

Chain OUTPUT (policy DROP 12 packets, 796 bytes)
pkts bytes target prot opt in out source destination
46 8392 ACCEPT all -- * lo 0.0.0.0/0 0.0.0.0/0
486 52750 ACCEPT all -- * * 0.0.0.0/0 0.0.0.0/0 state RELATED,ESTABLISHED
1 84 ACCEPT icmp -- * * 0.0.0.0/0 10.0.0.0/24 icmptype 8 limit: avg 4/sec burst 4
1 84 ACCEPT icmp -- * * 0.0.0.0/0 10.0.2.4 icmptype 8 limit: avg 4/sec burst 4
4 240 ACCEPT tcp -- * * 0.0.0.0/0 10.0.3.3 tcp dpt:22
```

## 6. Conclusão

A firewall foi configurada corretamente segundo uma política de default deny. A máquina SperServ só aceita o tráfego pedido no enunciado e bloqueia o restante por omissão. Os testes realizados confirmam o funcionamento esperado.